Astronomical Image Analysis: Photometry of NN Serpentis

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April 2026

Abstract

Imaging and time-series photometric data were analysed to investigate both extended and variable astronomical sources. Multi-filter observations of the spiral galaxy M91 were combined to produce a colour composite image, revealing the spatial distribution of stellar populations and regions of ionised gas associated with ongoing star formation. Aperture photometry was performed on the eclipsing binary NN Serpentis to measure its brightness and construct a lightcurve.

Ground-based observations from the IAC80 telescope show no statistically significant variability over the ∼0.5 hour observing window, consistent with incomplete phase coverage relative to the orbital period. In contrast, analysis of TESS photometry using a Lomb–Scargle periodogram reveals a strong periodic signal with a period of $P = 0.13007$ days. The corresponding phase-folded lightcurve exhibits a well-defined eclipse, confirming the eclipsing nature of the system.

These results demonstrate the importance of wavelength-dependent imaging for probing galactic structure and highlight the necessity of continuous, high-cadence observations for reliably detecting periodic variability.

1 Introduction

Astronomical imaging and photometry provide essential tools for probing both the spatial structure and temporal behaviour of astrophysical systems. In particular, multi-wavelength imaging allows different physical components of galaxies to be isolated, while time-resolved photometry enables the detection and characterisation of variable sources such as eclipsing binaries.

In this work, imaging data from the IAC80 telescope are analysed to investigate two distinct problems. First, multi-filter observations of the spiral galaxy M91 are used to construct a colour composite image and examine the spatial distribution of stellar populations and star-forming regions. Second, aperture photometry is applied to the eclipsing binary NN Serpentis in order to measure its brightness and construct a lightcurve.

Due to the limited temporal coverage of the ground-based observations, complementary photometric data from the Transiting Exoplanet Survey Satellite (TESS) are analysed to determine the orbital period of the system. This combined approach allows both the limitations of ground-based observations and the advantages of space-based photometry to be assessed within a consistent framework.

2 Theory

2.1 CCD Image Reduction

Astronomical images are recorded using charge-coupled devices (CCDs), which measure incoming photons as counts. Raw CCD images contain instrumental effects, including an electronic bias offset and pixel-to-pixel sensitivity variations.

The bias level is removed by subtracting a master bias frame constructed from multiple bias exposures. Flat-field images are normally used to correct for variations in detector response;

however, in this analysis flat-field correction was not applied to the clear-filter images, and residual sensitivity variations are therefore expected to introduce a small systematic uncertainty.

2.2 Aperture Photometry

Aperture photometry estimates the flux of a source by summing pixel values within a defined aperture and subtracting a local background contribution. The accuracy of this method depends on the choice of aperture size, which must balance inclusion of the point spread function (PSF) against contamination from background noise. In practice, the optimal aperture depends on the seeing conditions and detector characteristics, and suboptimal choices can introduce systematic errors in the measured flux.

$$F = F_{\text{aperture}} - F_{\text{background}}. \tag{1}$$

2.3 Photometric Calibration

Astronomical brightness is expressed on a logarithmic magnitude scale. Instrumental magnitudes derived from measured fluxes are converted to calibrated magnitudes using a zeropoint:

$$m = -2.5 \log_{10}(F) + ZP. \tag{2}$$

The zeropoint is determined using comparison stars of known magnitude, allowing all measured fluxes to be placed on a consistent photometric scale.

2.4 Time-Series Analysis

Periodic variability in astronomical sources can be identified using a Lomb–Scargle periodogram, which is well suited to unevenly sampled data. The dominant frequency in the periodogram corresponds to the most significant periodic signal, and the associated period is given by $P = 1/f$.

The lightcurve can then be phase-folded using this period to reveal repeating features such as eclipses, allowing the structure of the variability to be analysed.

3 Methodology

3.1 Data Reduction

Standard CCD reduction techniques were applied, including bias subtraction using a master bias frame constructed from multiple exposures. Flat-field correction was not applied to the clear-filter images used for NN Ser photometry. While bias subtraction removes the dominant electronic offset, the omission of flat-fielding introduces a potential systematic uncertainty due to pixel-to-pixel sensitivity variations.

This effect is expected to be small over the limited spatial region used for aperture photometry; however, it remains a limitation of the reduction process and may contribute to residual photometric scatter.

3.2 Initial Photometric Setup

Aperture photometry was performed using a fixed aperture radius of 6 pixels. This choice represents a compromise between maximising signal and limiting background contamination, and was adopted as a practical aperture for the dataset rather than one explicitly optimised through a curve-of-growth analysis. A more rigorous approach would involve such an analysis to verify that a consistent fraction of the PSF is captured across all images.

Photometric calibration was achieved using comparison stars with known magnitudes. The zeropoint was calculated for each image and averaged. Stars that introduced significant scatter were excluded, improving internal consistency. However, this procedure does not account for systematic uncertainties such as catalogue errors or colour-dependent effects.

3.3 Image Stacking and Colour Composite

For M91, reduced images in the B, V, and Hα filters were aligned and combined to produce stacked images in each band. These stacked images were then used to construct an RGB composite image, with B mapped to blue, V to green, and Hα to red. Prior to combination, each image was scaled to enhance contrast and ensure a comparable dynamic range between filters.

3.4 Source Detection and Identification

Sources in the NN Ser images were detected using the `DAOStarFinder` algorithm, with detection thresholds defined relative to the background noise level. The target (NN Ser) and nearby comparison stars were identified by matching the detected source positions to a provided finder chart.

The identification was based on the relative geometry of the field, including the separation and orientation of nearby stars. Only stars that appeared isolated and showed consistent behaviour across images were selected as comparison stars for photometric calibration.

3.5 Photometry and Calibration

Aperture photometry was performed on NN Ser and the selected comparison stars using circular apertures of radius 6 pixels. The local background level was estimated using an annulus with inner and outer radii of 10 and 15 pixels, respectively, and subtracted from the aperture flux to obtain the net source flux.

Instrumental magnitudes were calculated from the measured fluxes and converted to calibrated magnitudes using a zeropoint derived from the comparison stars. The zeropoint was determined for each image and averaged to obtain a stable calibration. Comparison stars that introduced significant scatter in the zeropoint were excluded to improve the reliability of the calibration.

3.6 Lightcurve Construction

Observation times were extracted from the FITS headers and converted to Julian Date. The calibrated magnitude of NN Ser was plotted as a function of time to construct a lightcurve. The time axis was expressed relative to the first observation to highlight short-term variability.

3.7 TESS Time-Series Analysis

TESS photometric data were analysed to determine the orbital period of NN Ser. The lightcurve was first cleaned by removing non-physical values and normalised by the median flux. A Lomb–Scargle periodogram was then computed to identify periodic signals in the data.

The period corresponding to the maximum power in the periodogram was adopted as the best estimate of the orbital period. The lightcurve was subsequently phase-folded using this period, and multiple cycles were plotted to improve visualisation of the eclipse structure.

4 Results

4.1 M91 Stacked Imaging

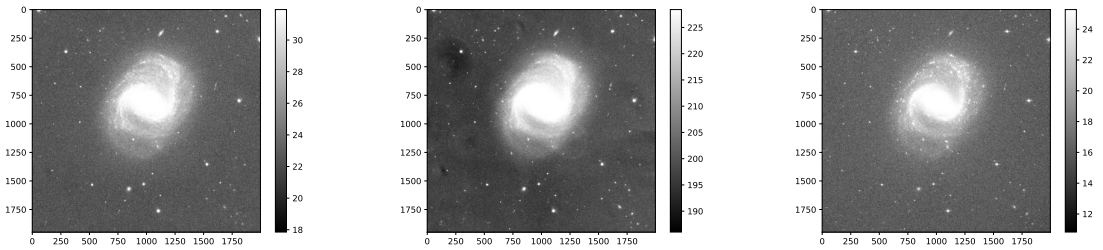


Figure 1: Stacked images of M91 in the B (left), V (centre), and Hα (right) filters. The B-band image highlights extended spiral structure and regions dominated by young stellar populations. The V-band image shows a smoother light distribution dominated by the central bulge and older stellar population. The Hα image reveals clumpy, localised emission associated with regions of active star formation along the spiral arms.

The stacked images show clear structural differences between filters, reflecting the underlying physical processes within the galaxy. The B-band image exhibits enhanced emission along the spiral arms, tracing young, massive stars and recent star formation. In contrast, the V-band image is dominated by the central bulge and provides a smoother representation of the overall stellar distribution.

The Hα image displays patchy, high-contrast emission regions concentrated along the spiral arms. These regions correspond to ionised hydrogen gas, indicating ongoing star formation in discrete locations rather than uniformly across the galaxy. The spatial separation between the smooth stellar component and the clumpy Hα emission demonstrates the value of multi-wavelength observations for separating distinct astrophysical components.

A residual detector artefact is visible at the edge of the images, appearing as a vertical strip of reduced sensitivity. This feature is consistent with structures observed in the flat-field images and arises due to non-uniform detector response. As flat-field correction was not applied to these images, this artefact remains present but does not significantly affect the interpretation of the galaxy structure, as the region of interest lies well within the central field.

4.2 M91 Colour Composite

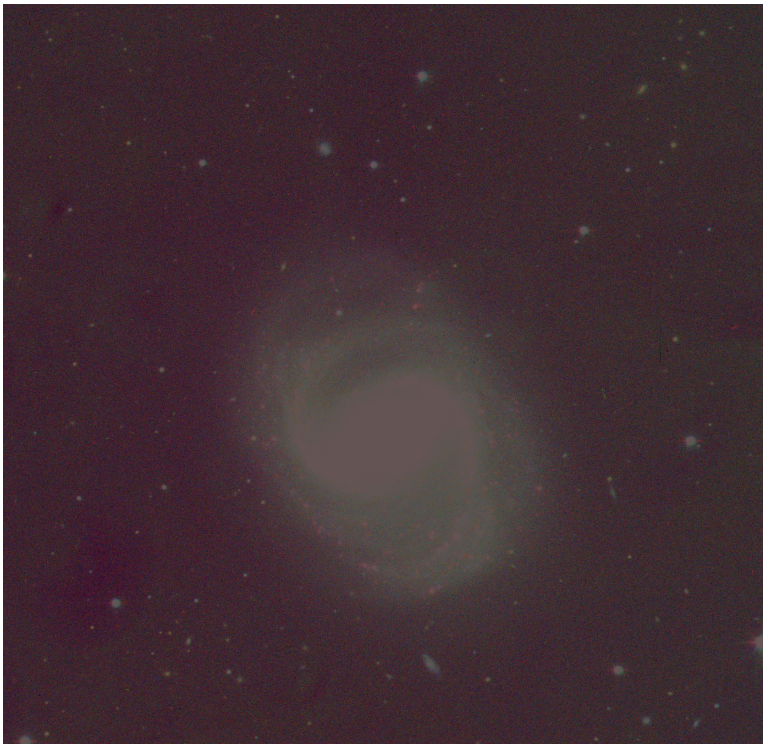


Figure 2: RGB composite image of M91 constructed from the stacked B (blue), V (green), and Hα (red) images. The central bulge appears bright and relatively smooth, while the spiral arms show enhanced structure. Localised red regions correspond to Hα emission, indicating active star formation within the spiral arms.

The RGB composite image highlights the different physical components of the galaxy. The central region is dominated by the older stellar population, producing a smooth and bright appearance, while the spiral arms exhibit greater structure and colour variation. Localised red emission regions trace Hα emission and therefore ongoing star formation, consistent with the interpretation from the individual filter images.

4.3 Source Identification

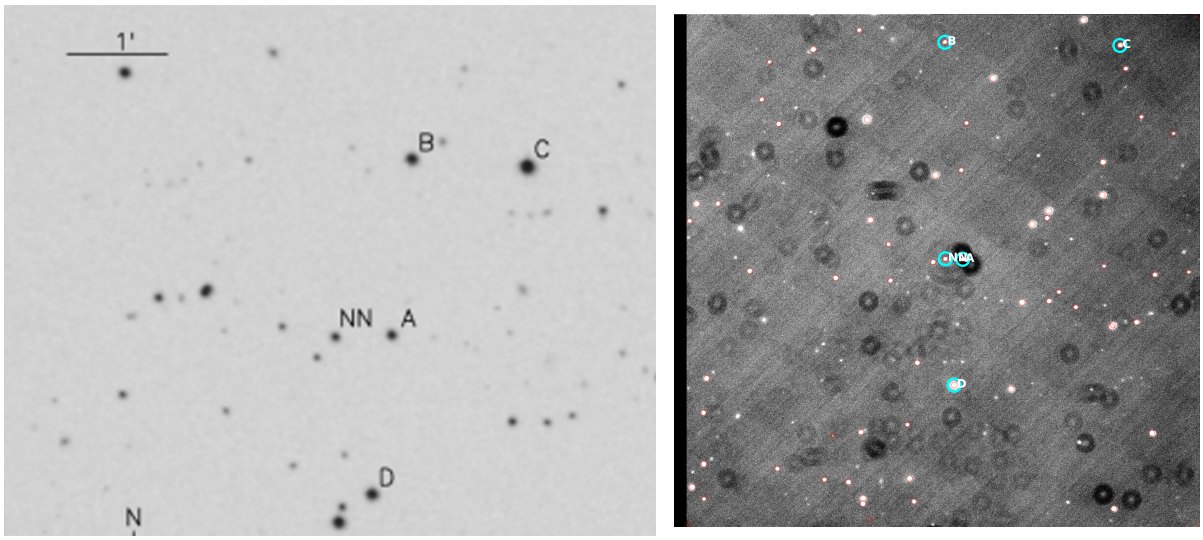


Figure 3: Comparison between the provided finder chart (left) and the reduced NN Ser image with detected sources and identified stars highlighted (right). The target (NN) and comparison stars (A, B, C, and D) were identified by matching their relative positions and separations.

The identification of NN Serpentis and the comparison stars was carried out by matching the detected source positions to the provided finder chart (Figure 3). The relative geometry of the field, including the separation and orientation of nearby stars, was used to distinguish the target and the local photometric sequence. The agreement between the detected source positions and the finder chart supports the identification adopted for the photometric analysis.

4.4 NN Ser Photometry

Aperture photometry was performed on NN Ser and nearby comparison stars, and calibrated magnitudes were obtained using a zeropoint derived from the selected comparison stars. The zeropoint values show good consistency, with a mean value of

$$ZP = 26.22 \pm 0.01 \text{ mag}. \tag{3}$$

The calibrated magnitude of NN Ser was found to be

$$m_{\mathrm{NN}} = 15.03 \text{ mag}, \tag{4}$$

while the calibrated magnitude of star A was

$$m_{\mathrm{A}} = 13.81 \text{ mag}. \tag{5}$$

The small zeropoint scatter indicates good internal consistency in the calibration. However, this uncertainty reflects the scatter among the selected comparison stars and is likely to underestimate the full photometric uncertainty, which may also include systematic contributions from aperture choice, catalogue magnitudes, and the omission of flat-field correction.

4.5 IAC80 Lightcurve

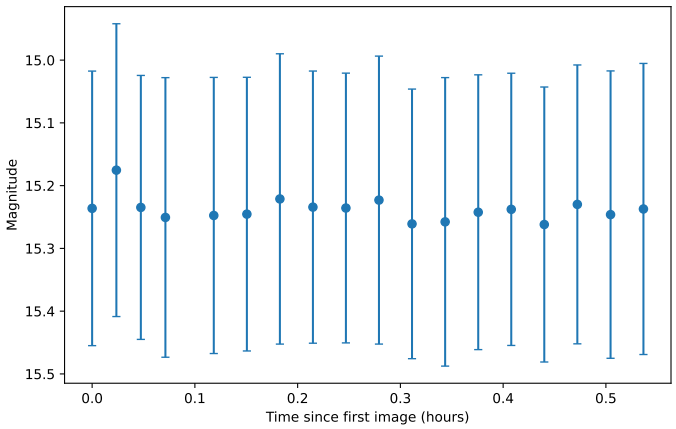


Figure 4: Lightcurve of NN Ser obtained from IAC80 observations, showing calibrated magnitude as a function of time since the first exposure.

The IAC80 lightcurve shows no significant variation in brightness over the duration of the observations, with the magnitude remaining approximately constant within the measurement uncertainties. The observed scatter is consistent with photometric noise and calibration uncertainty.

This result is expected given the short observational baseline of approximately 0.5 hours, which samples only a small fraction of the orbital period of NN Ser ($\sim$3.1 hours, corresponding to $P = 0.13007$ days). It is therefore likely that the system was observed entirely outside eclipse, and the absence of a clear dip in brightness is consistent with this phase coverage.

4.6 TESS Period Analysis

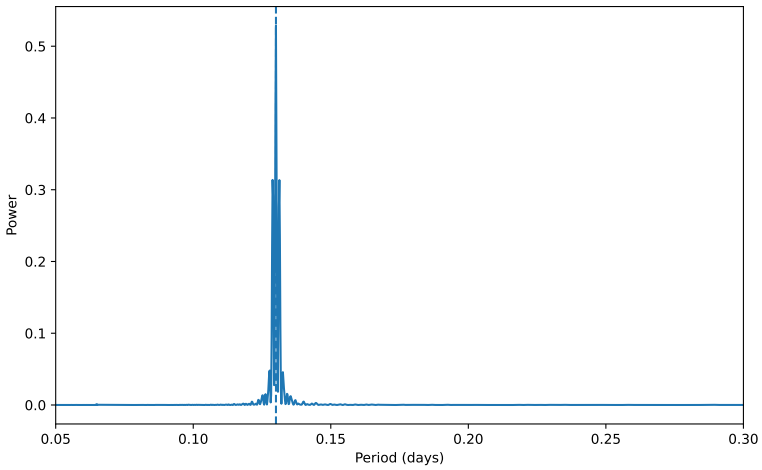


Figure 5: Lomb–Scargle periodogram of NN Ser derived from TESS data. A strong peak is observed at a period of $P \approx 0.13$ days.

The Lomb–Scargle periodogram exhibits a strong and well-defined peak at $P \approx 0.13$ days, indicating a stable periodic signal. The sharpness and isolation of the peak suggest that the period is well constrained by the data and not strongly affected by noise or aliasing.

This result is consistent with expectations for NN Ser and confirms that the dominant variability arises from orbital modulation of the binary system.

4.7 Phase-Folded Lightcurve

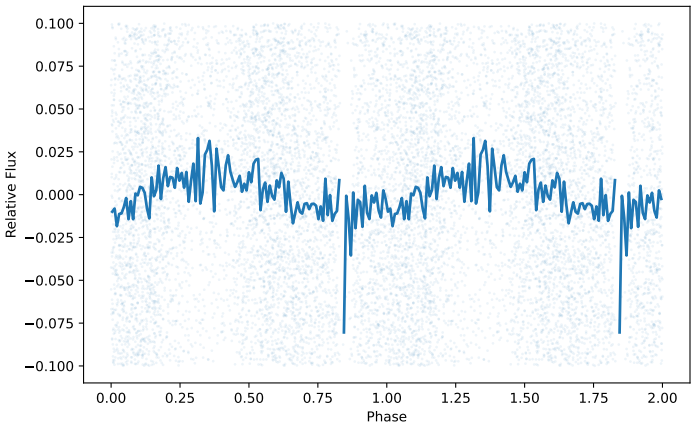


Figure 6: Phase-folded lightcurve of NN Ser using a period of $P = 0.13007$ days. Two cycles are shown for clarity.

The phase-folded lightcurve reveals a clear and repeatable eclipse feature, characterised by a sharp decrease in flux at a well-defined phase. The eclipse appears narrow and approximately symmetric, consistent with a compact eclipsing binary system.

The presence of a single dominant eclipse indicates that the observed variability is driven by orbital motion. The depth and shape of the eclipse suggest a high inclination angle, such that one component passes directly in front of the other along the line of sight.

The consistency of the eclipse across multiple cycles supports the reliability of the derived period and is consistent with the established classification of NN Ser as a short-period eclipsing binary system.

5 Discussion

5.1 M91: Multi-Wavelength Structure

The multi-filter imaging of M91 illustrates how observations at different wavelengths trace distinct physical components of a galaxy. The B-band emission is dominated by short-lived, massive stars, and therefore highlights regions of recent star formation concentrated along the spiral arms. In contrast, the V-band provides a smoother distribution, reflecting the older, more spatially extended stellar population that dominates the galactic bulge.

The Hα emission further isolates regions of ionised hydrogen, providing a direct tracer of ongoing star formation. The clumpy and localised morphology observed in the Hα image indicates that star formation occurs in discrete regions rather than uniformly across the disk. The spatial offset between the smooth stellar continuum and the Hα emission is consistent with spiral density wave theory, in which gas compression in the spiral arms triggers star formation.

A residual detector artefact is visible at the edge of the field, attributable to pixel-to-pixel sensitivity variations. As flat-field correction was not applied, this feature remains in the reduced images. While it does not affect the central region of interest, its presence highlights a systematic limitation of the reduction process. In a more precise analysis, flat-fielding would be required to ensure a uniform photometric response across the detector.

5.2 Photometric Calibration and Source Selection

The photometric calibration relies critically on the selection of stable comparison stars. Initial inclusion of an additional comparison star resulted in a significant increase in zeropoint scatter, indicating variability or systematic issues with that source. Restricting the calibration to a subset of consistent stars reduced the zeropoint uncertainty to approximately 0.01 mag, demonstrating the sensitivity of the calibration to reference star selection.

However, while the low scatter suggests good internal consistency, it likely underestimates the true uncertainty. Systematic effects such as catalogue magnitude errors, colour differences between target and comparison stars, and residual flat-field variations are not captured in the quoted zeropoint uncertainty. As a result, the reported precision should be interpreted as a lower bound rather than a complete error estimate.

The choice of aperture radius (6 pixels) represents a trade-off between maximising signal and minimising background noise. Although this aperture was not explicitly optimised, it provides a consistent basis for comparing the target and reference stars across the dataset. A more rigorous treatment would involve a curve-of-growth analysis to test whether the chosen aperture captures a stable fraction of the PSF in all images.

5.3 IAC80 Lightcurve and Observational Constraints

The IAC80 lightcurve shows no statistically significant variability, with observed scatter consistent with photometric uncertainties. Given the orbital period of NN Ser (~ 3.1 hours), the observational window of approximately 0.5 hours samples only about 16% of the orbital cycle. The absence of an eclipse in the dataset is therefore expected and does not imply a lack of intrinsic variability.

This highlights a key limitation of ground-based observations: restricted temporal coverage can severely limit sensitivity to periodic phenomena. In addition, atmospheric effects such as seeing variations and transparency fluctuations introduce additional noise that further reduces the detectability of small-amplitude variability.

A more robust detection of variability would require either longer continuous observations or coordinated observations across multiple sites to improve phase coverage.

5.4 TESS Period Determination and Eclipse Detection

The TESS dataset provides continuous, high-precision photometry, enabling a more reliable analysis of periodic variability. The Lomb–Scargle periodogram exhibits a strong and well-defined peak at approximately 0.13007 days, indicating a stable and coherent periodic signal.

The phase-folded lightcurve reveals a sharp and repeatable eclipse feature, consistent with a high-inclination eclipsing binary system. The narrow eclipse profile suggests that the emitting regions are compact and that the orbital geometry produces a near-occultation during eclipse.

While the period determination is robust, potential sources of uncertainty include aliasing and window-function effects, although these are reduced by the continuous nature of the TESS observations. The consistency of the signal across multiple cycles supports the reliability of the derived period.

5.5 Comparison of Ground-Based and Space-Based Observations

The combined use of IAC80 and TESS data highlights the complementary nature of ground-based and space-based observations. Ground-based imaging provides spatial resolution and flexibility in observational setup, but is limited by atmospheric conditions and restricted observing windows. In contrast, space-based missions such as TESS deliver continuous, high-precision photometry that is well suited to detecting and characterising periodic variability.

The absence of variability in the IAC80 data, compared with the clear periodic signal in the TESS data, demonstrates that observational limitations rather than intrinsic properties of the system dominate the ground-based result. This underscores the importance of matching observational strategy to the physical timescales of the system under study.

5.6 Uncertainty and Limitations

Several sources of uncertainty affect the results of this analysis. Random uncertainties arise from photon noise, background estimation, and detector readout noise, while systematic uncertainties are introduced through the omission of flat-field correction, the choice of comparison stars, and potential calibration biases.

The zeropoint scatter provides a measure of internal consistency but does not fully account for systematic effects. Additionally, uncertainties in aperture selection and background estimation may introduce small but non-negligible biases in the measured fluxes.

The IAC80 lightcurve is fundamentally limited by its short temporal baseline, preventing full sampling of the orbital period. In contrast, the TESS data significantly reduce both statistical and temporal uncertainties, allowing a more robust characterisation of the system.

Overall, while these uncertainties affect the precision of the measurements, they do not alter the primary conclusion that NN Ser is a short-period eclipsing binary system with a well-defined orbital period.

6 Conclusion

This analysis demonstrates the application of CCD image reduction, aperture photometry, and time-series analysis to both extended and variable astronomical sources. Multi-filter imaging of M91 reveals clear distinctions between stellar populations and regions of active star formation, illustrating the value of wavelength-dependent observations for probing galactic structure.

Photometric analysis of NN Ser yielded a calibrated magnitude of $m_{\mathrm{NN}} = 15.03$ mag, with a stable zeropoint indicating internally consistent calibration. The IAC80 lightcurve shows no detectable variability, consistent with the limited temporal coverage relative to the orbital period.

In contrast, TESS data provide a robust determination of the orbital period, $P = 0.13007$ days, and reveal a clear eclipse signature in the phase-folded lightcurve. This highlights the importance of continuous, high-cadence observations for detecting and characterising periodic variability.

Overall, the results provide a consistent picture of NN Ser as a short-period eclipsing binary system and demonstrate the complementary roles of ground-based and space-based observations.

References

[1] M. Fraser, *Astronomical Image Analysis Lab Manual*, UCD Physics (2024).

A Notebook Appendix

The full Jupyter notebook used for the data reduction, photometric analysis, period determination, and phase-folding is included in the following pages.

Extract data from zipped file

In [2]:
```
import zipfile

with zipfile.ZipFile("Astro_analysis_data.zip", 'r') as zip_ref:
    zip_ref.extractall("data")
```

In [181…
```
import glob
from astropy.io import fits
import numpy as np
import matplotlib.pyplot as plt
from astropy.stats import mad_std
!pip install photutils
from photutils.detection import DAOStarFinder
from photutils.aperture import CircularAperture, CircularAnnulus, aperture_photo
```

```
Requirement already satisfied: photutils in c:\users\lewis\anaconda3\lib\site-pac
kages (2.3.0)
Requirement already satisfied: numpy>=1.25 in c:\users\lewis\anaconda3\lib\site-p
ackages (from photutils) (2.1.3)
Requirement already satisfied: astropy>=5.3 in c:\users\lewis\anaconda3\lib\site-
packages (from photutils) (7.0.0)
Requirement already satisfied: scipy>=1.11.1 in c:\users\lewis\anaconda3\lib\site
-packages (from photutils) (1.15.3)
Requirement already satisfied: pyerfa>=2.0.1.1 in c:\users\lewis\anaconda3\lib\si
te-packages (from astropy>=5.3->photutils) (2.0.1.5)
Requirement already satisfied: astropy-iers-data>=0.2024.10.28.0.34.7 in c:\users
\lewis\anaconda3\lib\site-packages (from astropy>=5.3->photutils) (0.2025.1.13.0.
34.51)
Requirement already satisfied: PyYAML>=6.0.0 in c:\users\lewis\anaconda3\lib\site
-packages (from astropy>=5.3->photutils) (6.0.2)
Requirement already satisfied: packaging>=22.0.0 in c:\users\lewis\anaconda3\lib
\site-packages (from astropy>=5.3->photutils) (24.2)
```

Create Master Bias

In [3]:
```
# --- FIND BIAS FILES ---
bias_files = glob.glob("data/**/*.fit", recursive=True) + glob.glob("data/**/*.f
bias_files = [f for f in bias_files if "bias" in f.lower()]

print("Number of bias files:", len(bias_files))
print(bias_files[:10])

# --- LOAD AND STACK ---
bias_stack = []
for fname in bias_files:
    data = fits.open(fname)[0].data
    bias_stack.append(data)

bias_stack = np.array(bias_stack)

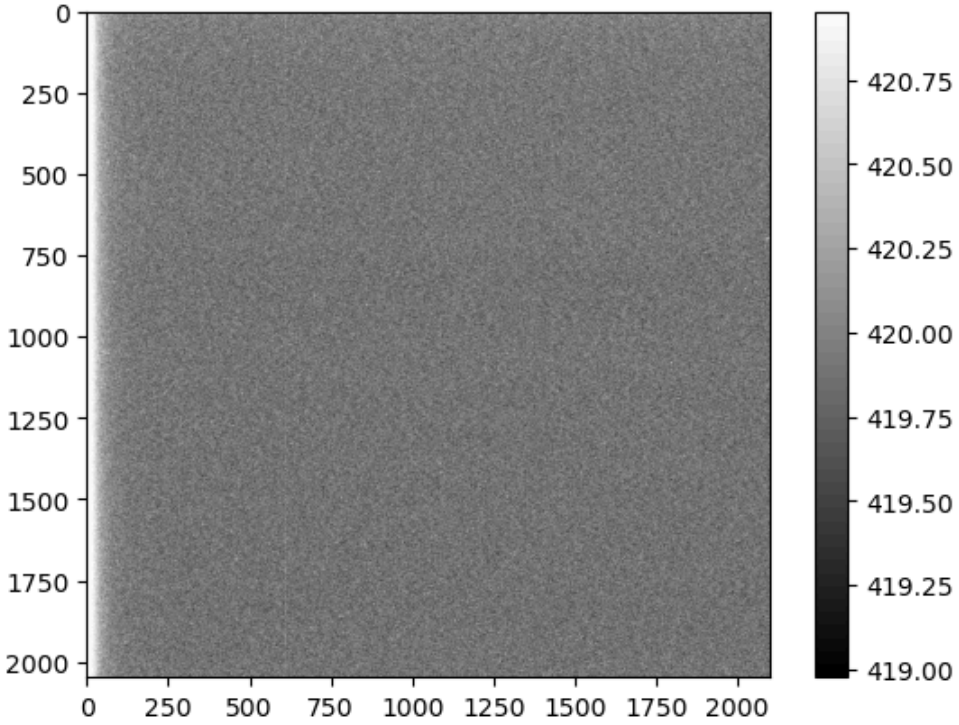
# --- CREATE MASTER BIAS ---
masterbias = np.mean(bias_stack, axis=0)
```

```
# --- PLOT (with proper contrast) ---
plt.imshow(masterbias, cmap='gray',
           vmin=np.percentile(masterbias, 5),
           vmax=np.percentile(masterbias, 95))
plt.colorbar()
plt.show()

# --- CHECK VALUES ---
print("Min:", masterbias.min())
print("Max:", masterbias.max())
print("Mean:", masterbias.mean())

# --- SAVE FILE ---
fits.writeto("masterbias.fits", masterbias, overwrite=True)
```

```
Number of bias files: 42
['data\\Astro_analysis_data\\IAC80_M91\\bias\\B130311_0001.fits', 'data\\Astro_an
alysis_data\\IAC80_M91\\bias\\B130311_0002.fits', 'data\\Astro_analysis_data\\IAC
80_M91\\bias\\B130311_0003.fits', 'data\\Astro_analysis_data\\IAC80_M91\\bias\\B1
30311_0004.fits', 'data\\Astro_analysis_data\\IAC80_M91\\bias\\B130311_0005.fit
s', 'data\\Astro_analysis_data\\IAC80_M91\\bias\\B130311_0006.fits', 'data\\Astro
_analysis_data\\IAC80_M91\\bias\\B130311_0007.fits', 'data\\Astro_analysis_data
\\IAC80_M91\\bias\\B130311_0008.fits', 'data\\Astro_analysis_data\\IAC80_M91\\bia
s\\B130311_0009.fits', 'data\\Astro_analysis_data\\IAC80_M91\\bias\\B130311_0010.
fits']
```




```
Min: 415.0238095238095
Max: 2318.4523809523807
Mean: 420.0347484457193
```

Create Master Flats of B, V, ha

In [4]:
```
# --- FIND ALL FLAT FILES ---
flat_files = glob.glob("data/**/*.fit", recursive=True) + glob.glob("data/**/*.f
```

```
flat_files = [f for f in flat_files if "flat" in f.lower()]

print("Number of flat files:", len(flat_files))
print(flat_files[:20])

# --- GROUP BY FILTER ---
def normpath(p):
    return p.replace("\\", "/").lower()

B_flats  = [f for f in flat_files if "/b/" in normpath(f)]
V_flats  = [f for f in flat_files if "/v/" in normpath(f)]
Ha_flats = [f for f in flat_files if "ha" in normpath(f) or "halpha" in normpath

print("B flats:", len(B_flats))
print("V flats:", len(V_flats))
print("Halpha flats:", len(Ha_flats))

# --- FUNCTION TO MAKE NORMALIZED MASTER FLAT ---
def make_normalized_masterflat(file_list, masterbias):
    stack = []
    for fname in file_list:
        data = fits.open(fname)[0].data
        bias_sub = data - masterbias
        stack.append(bias_sub)

    stack = np.array(stack)
    masterflat = np.mean(stack, axis=0)

    # normalize using a central 100x100 region
    ny, nx = masterflat.shape
    y0, y1 = ny // 2 - 50, ny // 2 + 50
    x0, x1 = nx // 2 - 50, nx // 2 + 50
    center_mean = np.mean(masterflat[y0:y1, x0:x1])

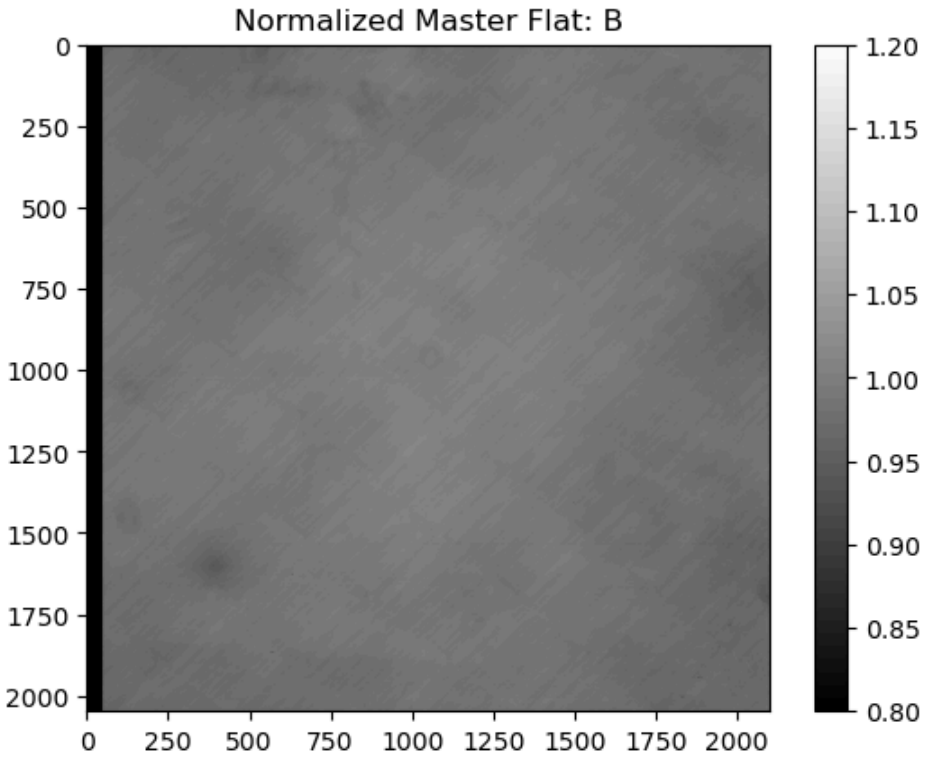
    norm_masterflat = masterflat / center_mean
    return norm_masterflat

# --- MAKE MASTER FLATS ---
B_masterflat = make_normalized_masterflat(B_flats, masterbias)
V_masterflat = make_normalized_masterflat(V_flats, masterbias)
Ha_masterflat = make_normalized_masterflat(Ha_flats, masterbias)

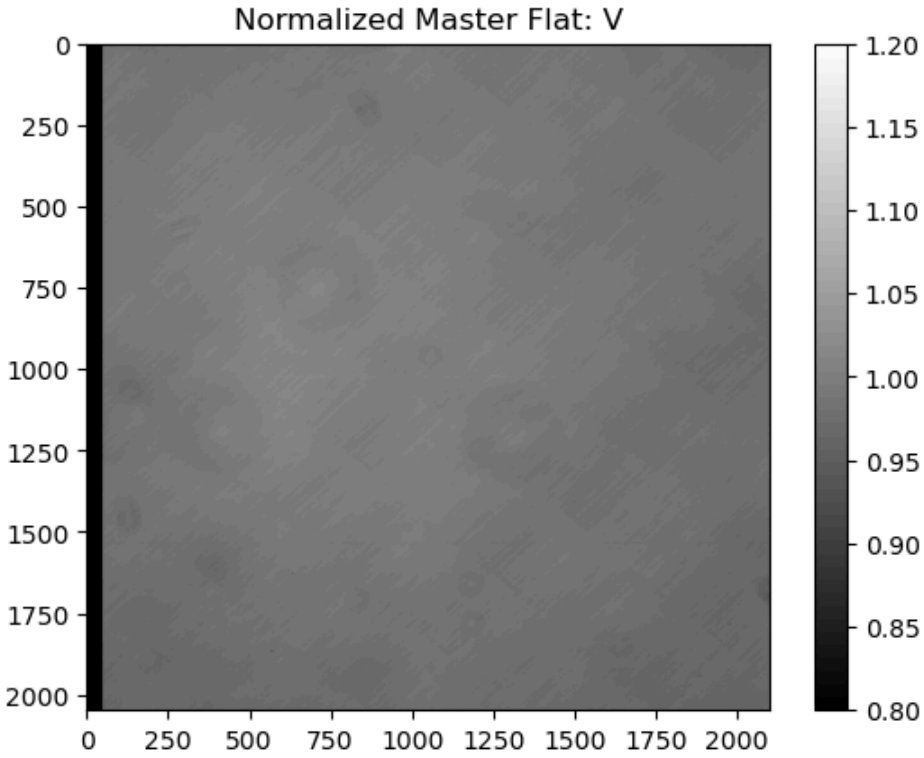
# --- PLOT RESULTS ---
for flat, name in [(B_masterflat, "B"), (V_masterflat, "V"), (Ha_masterflat, "Ha
    plt.figure()
    plt.imshow(flat, cmap="gray", vmin=0.8, vmax=1.2)
    plt.colorbar()
    plt.title(f"Normalized Master Flat: {name}")
    plt.show()
    print(name, "min/max/mean =", flat.min(), flat.max(), flat.mean())

# --- SAVE FILES ---
fits.writeto("nmasterflat_B.fits", B_masterflat, overwrite=True)
fits.writeto("nmasterflat_V.fits", V_masterflat, overwrite=True)
fits.writeto("nmasterflat_Ha.fits", Ha_masterflat, overwrite=True)
```

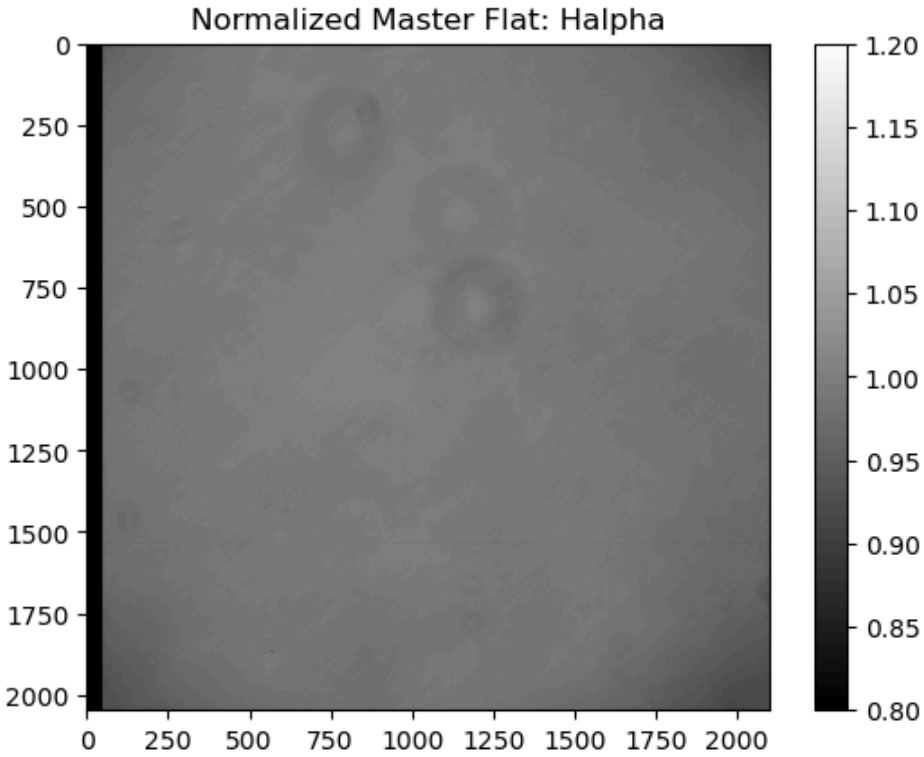
```
Number of flat files: 40
['data\\Astro_analysis_data\\IAC80_M91\\flats\\B\\F130311_0022.fits', 'data\\Astr
o_analysis_data\\IAC80_M91\\flats\\B\\F130311_0023.fits', 'data\\Astro_analysis_d
ata\\IAC80_M91\\flats\\B\\F130311_0024.fits', 'data\\Astro_analysis_data\\IAC80_M
91\\flats\\B\\F130311_0025.fits', 'data\\Astro_analysis_data\\IAC80_M91\\flats\\B
\\F130311_0026.fits', 'data\\Astro_analysis_data\\IAC80_M91\\flats\\B\\F130311_00
27.fits', 'data\\Astro_analysis_data\\IAC80_M91\\flats\\B\\F130311_0028.fits', 'd
ata\\Astro_analysis_data\\IAC80_M91\\flats\\B\\F130311_0029.fits', 'data\\Astro_a
nalysis_data\\IAC80_M91\\flats\\B\\F130311_0030.fits', 'data\\Astro_analysis_data
\\IAC80_M91\\flats\\B\\F130311_0031.fits', 'data\\Astro_analysis_data\\IAC80_M91
\\flats\\B\\F130311_0032.fits', 'data\\Astro_analysis_data\\IAC80_M91\\flats\\Hal
pha_NII\\F130311_0066.fits', 'data\\Astro_analysis_data\\IAC80_M91\\flats\\Halpha
_NII\\F130311_0067.fits', 'data\\Astro_analysis_data\\IAC80_M91\\flats\\Halpha_NI
I\\F130311_0068.fits', 'data\\Astro_analysis_data\\IAC80_M91\\flats\\Halpha_NII
\\F130311_0069.fits', 'data\\Astro_analysis_data\\IAC80_M91\\flats\\Halpha_NII\\F
130311_0070.fits', 'data\\Astro_analysis_data\\IAC80_M91\\flats\\Halpha_NII\\F130
311_0071.fits', 'data\\Astro_analysis_data\\IAC80_M91\\flats\\Halpha_NII\\F130311
_0072.fits', 'data\\Astro_analysis_data\\IAC80_M91\\flats\\Halpha_NII\\F130311_00
73.fits', 'data\\Astro_analysis_data\\IAC80_M91\\flats\\Halpha_NII\\F130311_0074.
fits']
B flats: 11
V flats: 11
Halpha flats: 11
```




```
B min/max/mean = -0.0002880572986345981 5 2.055638495077591 0.9629645576168221
```




```
V min/max/mean = -0.00028181147392484693 2.0554464203088183 0.9639597948041547
```




```
Halpha min/max/mean = -0.0002649054305945565 2.0390930758988963 0.962741057662018
3
```

Stack Filters

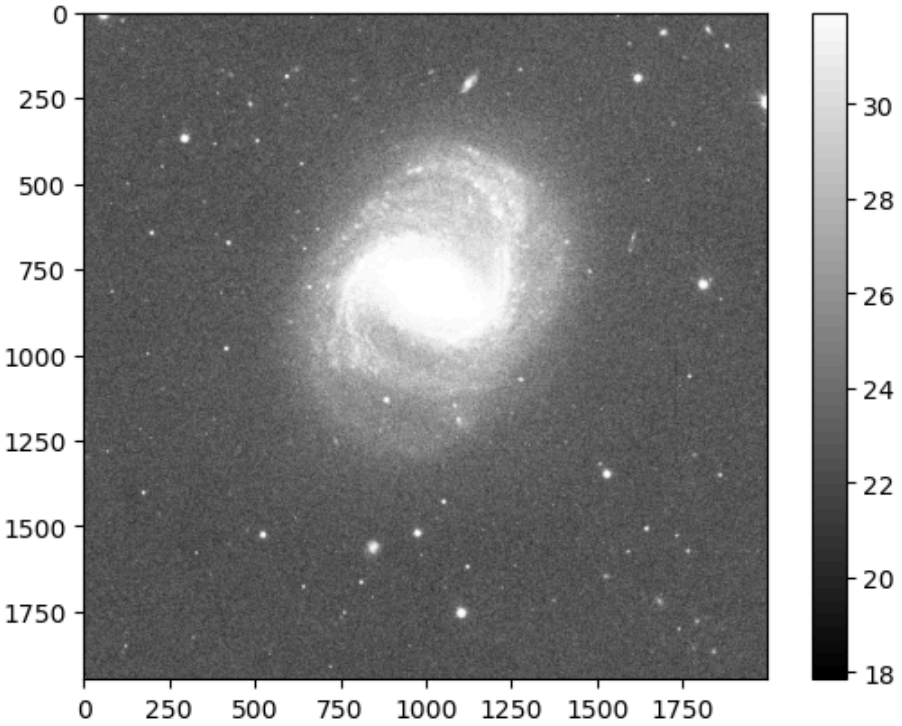
In [191…

```
# --- replace remaining NaNs with median of each image ---
B_stack_clean = np.where(np.isnan(B_stack), np.nanmedian(B_stack), B_stack)
V_stack_clean = np.where(np.isnan(V_stack), np.nanmedian(V_stack), V_stack)
Ha_stack_clean = np.where(np.isnan(Ha_stack), np.nanmedian(Ha_stack), Ha_stack)

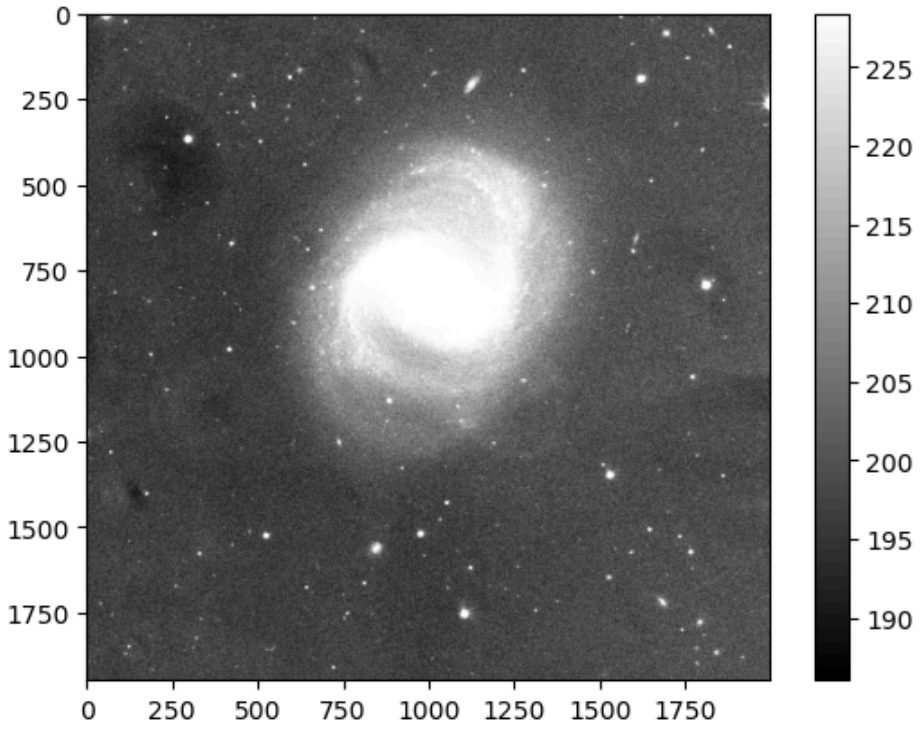
# --- crop off bad edges ---
crop = 50
B_crop = B_stack_clean[crop:-crop, crop:-crop]
V_crop = V_stack_clean[crop:-crop, crop:-crop]
Ha_crop = Ha_stack_clean[crop:-crop, crop:-crop]

# --- plot cleaned stacks ---
for img, name in [(B_crop, "B"), (V_crop, "V"), (Ha_crop, "Halpha")]:
    plt.figure()
    plt.imshow(img,
               cmap="gray",
               vmin=np.percentile(img, 5),
               vmax=np.percentile(img, 95))
    plt.colorbar()
    plt.savefig(f"Cleaned stacked {name} image")
    plt.show()

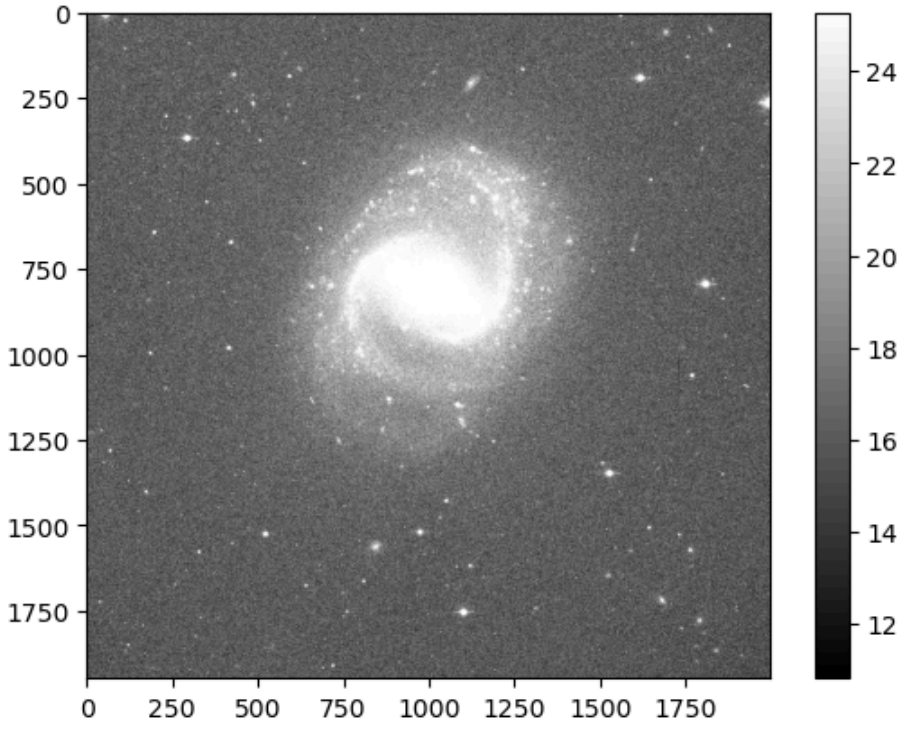
    print(name, "min/max/mean =", np.min(img), np.max(img), np.mean(img))
```




```
B min/max/mean = -207.57681462556283 14937.04304675085 24.46599034221436
```




```
V min/max/mean = -46.563814622263926 50859.2074287388 204.41638709355172
```




```
Halpha min/max/mean = -213.22905716452973 13416.927850476686 17.847469019443032
```

Colour M91

In [205…

```
def scale(img):
    vmin = np.percentile(img, 1)
    vmax = np.percentile(img, 95)
```

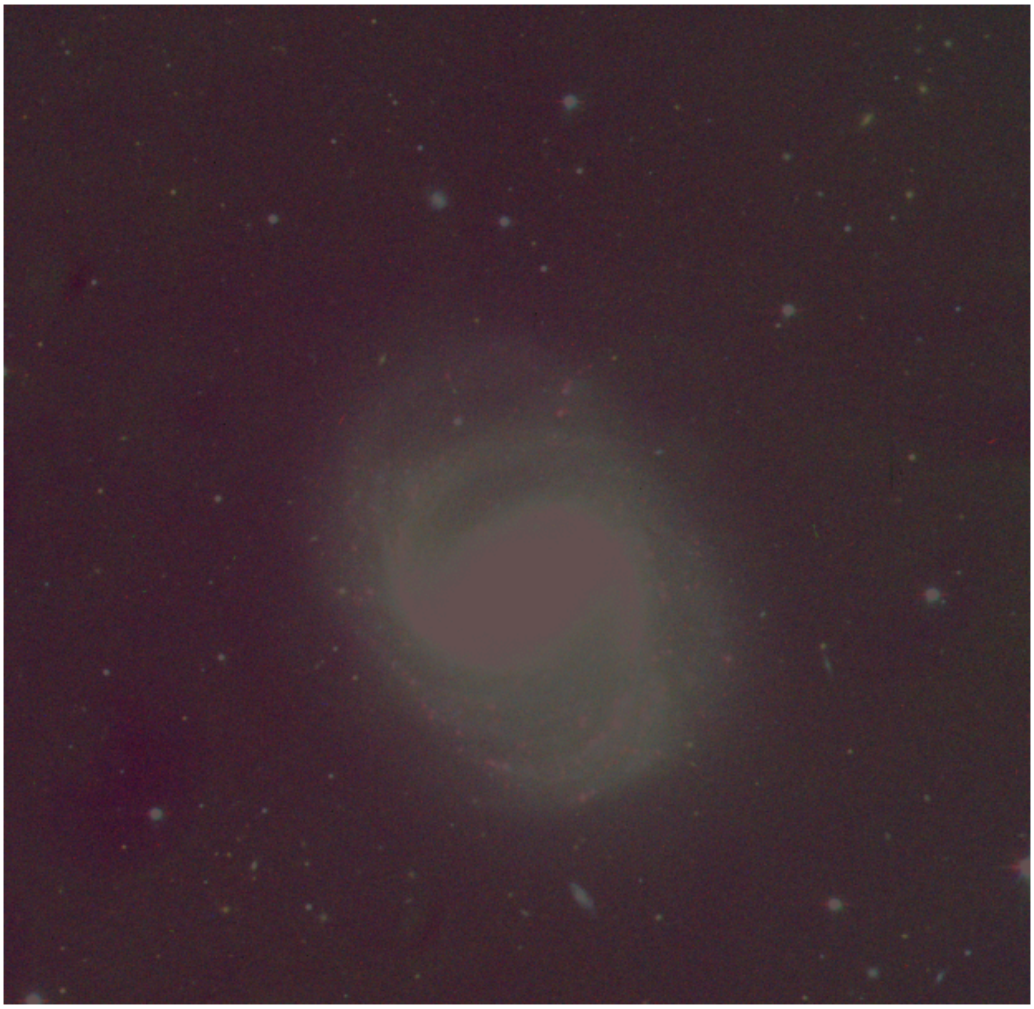
```
    img = (img - vmin) / (vmax - vmin)
    return np.clip(img, 0, 1)

B_scaled = scale(B_crop)
V_scaled = scale(V_crop)
Ha_scaled = scale(Ha_crop)

rgb = make_lupton_rgb(Ha_scaled*1.2, V_scaled, B_scaled, stretch=2, Q=5)

plt.figure(figsize=(8,8))
plt.imshow(rgb, origin="lower")
plt.axis("off")
plt.savefig("m91_rgb.pdf", bbox_inches="tight", dpi=300)
plt.show()
```

```
C:\Users\lewis\anaconda3\Lib\site-packages\astropy\visualization\lupton_rgb.py:305: RuntimeWarning: invalid value encountered in divide
  fInorm = np.where(Int <= 0, 0, np.true_divide(fI, Int))
```



Source Identification

In [199…

```
# --- find NN Ser science images ---
nn_files = glob.glob("data/**/*.fit", recursive=True) + glob.glob("data/**/*.fit
nn_files = [f for f in nn_files if "object" in f.lower() and ("nn" in f.lower()
```

```
print("Number of NN Ser science images:", len(nn_files))
print(nn_files[:10])

# --- reduce one NN Ser image ---
nn_test_file = nn_files[0]
nn_data = fits.open(nn_test_file)[0].data.astype(float)

# use the clear-filter images with the already made flat that matches them best
# if your NN Ser images are in a 'clear' folder, start by bias-subtracting only
nn_reduced = nn_data - masterbias

# --- display image ---
plt.figure(figsize=(8, 8))
plt.imshow(nn_reduced, cmap="gray",
           vmin=np.percentile(nn_reduced, 5),
           vmax=np.percentile(nn_reduced, 95))
plt.colorbar()
plt.title("NN Ser test image")
plt.show()

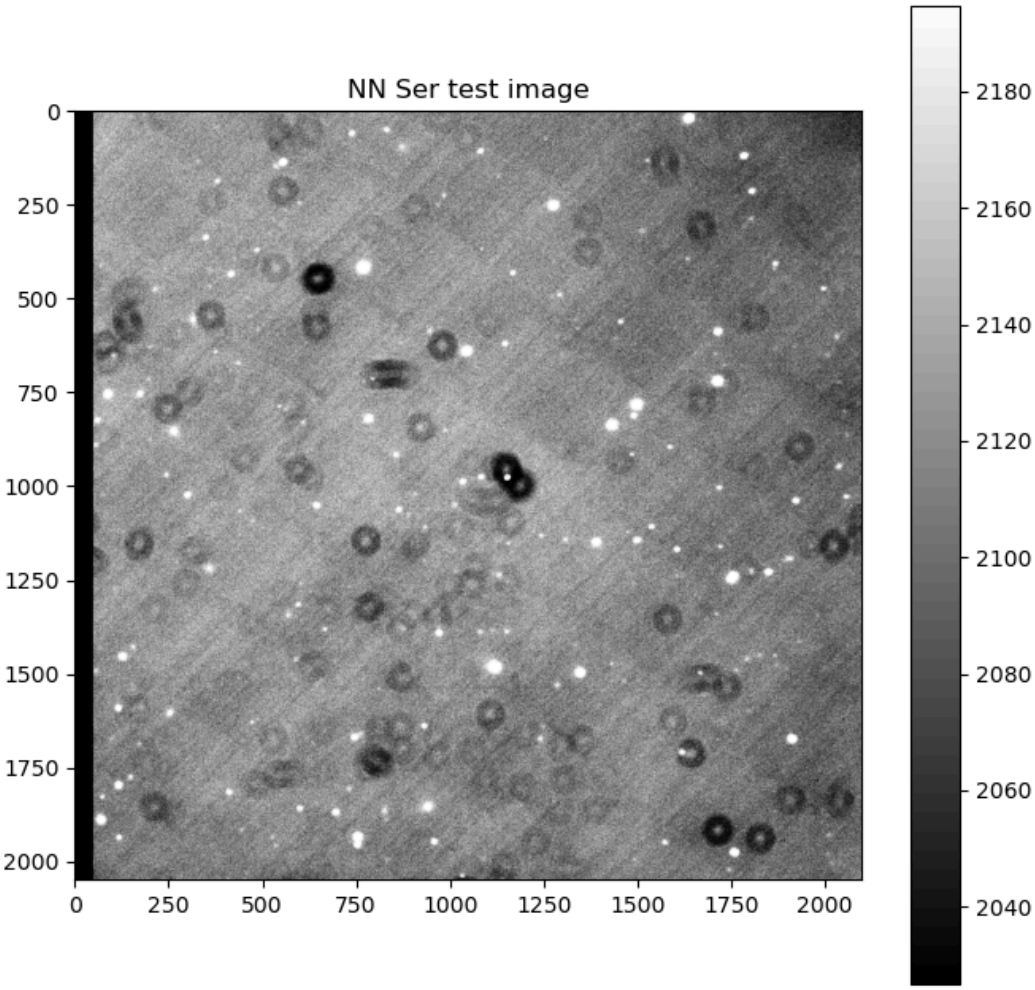
# --- detect stars ---
bkg_sigma = mad_std(nn_reduced)
daofind = DAOStarFinder(fwhm=6.0, threshold=5* bkg_sigma)
sources = daofind(nn_reduced)

print(sources)

# --- overplot detected sources ---
plt.figure(figsize=(8, 8))
plt.imshow(nn_reduced, cmap="gray",
           vmin=np.percentile(nn_reduced, 5),
           vmax=np.percentile(nn_reduced, 95))

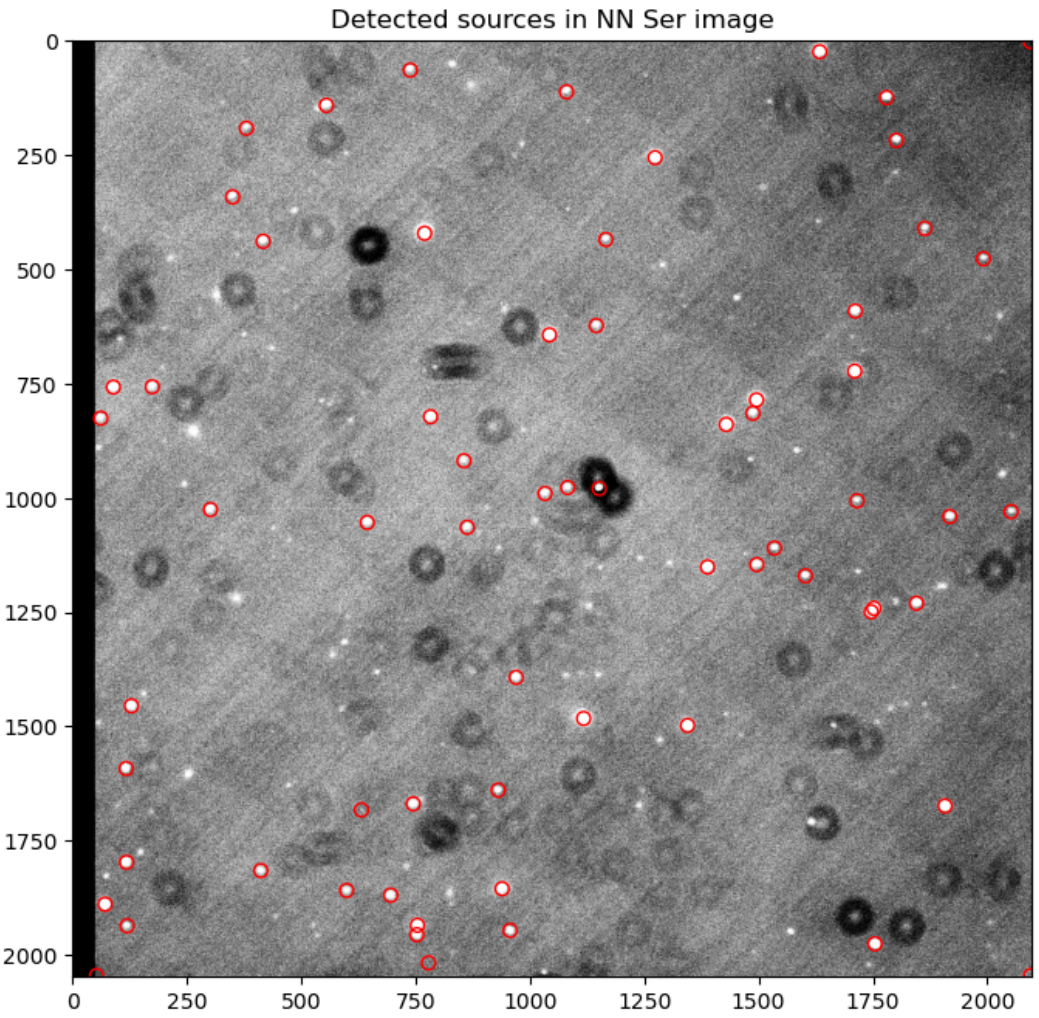
if sources is not None:
    plt.scatter(sources["xcentroid"], sources["ycentroid"],
                s=40, facecolors="none", edgecolors="red")
plt.title("Detected sources in NN Ser image")
plt.show()
```

```
Number of NN Ser science images: 36
['data\\Astro_analysis_data\\IAC80_NN_Ser\\nnser\\object\\clear\\O120308_0175.fit
s', 'data\\Astro_analysis_data\\IAC80_NN_Ser\\nnser\\object\\clear\\O120308_0176.
fits', 'data\\Astro_analysis_data\\IAC80_NN_Ser\\nnser\\object\\clear\\O120308_01
77.fits', 'data\\Astro_analysis_data\\IAC80_NN_Ser\\nnser\\object\\clear\\O120308
_0178.fits', 'data\\Astro_analysis_data\\IAC80_NN_Ser\\nnser\\object\\clear\\O120
308_0179.fits', 'data\\Astro_analysis_data\\IAC80_NN_Ser\\nnser\\object\\clear\\O
120308_0180.fits', 'data\\Astro_analysis_data\\IAC80_NN_Ser\\nnser\\object\\clear
\\O120308_0181.fits', 'data\\Astro_analysis_data\\IAC80_NN_Ser\\nnser\\object\\cl
ear\\O120308_0182.fits', 'data\\Astro_analysis_data\\IAC80_NN_Ser\\nnser\\object
\\clear\\O120308_0183.fits', 'data\\Astro_analysis_data\\IAC80_NN_Ser\\nnser\\obj
ect\\clear\\O120308_0184.fits']
```



id	xcentroid	...	mag	daofind_mag
1	2094.7288131815603	...	-11.762717203552576	-2.720258811279318
2	1633.6786564949032	...	-14.247575659212002	-4.225623004891699
3	738.1022557133788	...	-12.723273125801349	-0.9964024847801346
4	1079.9884476773912	...	-12.706688004404953	-0.9457374992883505
5	1779.6944403227665	...	-13.065166052269124	-2.3291746446714665
6	554.7624528024411	...	-13.258857922634121	-2.8538949328136147
7	379.94876689119707	...	-12.622961730610305	-0.002341553617028936
8	1800.9912851537983	...	-12.80140270980191	-1.3903097730637295
9	1273.572894451519	...	-14.185870646484307	-4.169867896302252
10	349.7652577445256	...	-12.725615942091705	-0.9535253622314976
...	...	...	...	...
57	695.4467794470346	...	-12.901585898609525	-1.5906756970954161
58	70.97014718701575	...	-13.73834957019871	-3.325868253947111
59	753.8194268355264	...	-13.85553262119997	-3.483246013666046
60	119.18140222768272	...	-12.664759407923246	-0.5787641427798623
61	956.5069070152504	...	-12.756256997928759	-0.995077694759491
62	752.7790399954393	...	-12.900598241439369	-1.5731358162805704
63	1754.3806231523902	...	-13.275259477143466	-2.3889318613873964
64	778.2765148875178	...	-12.545241735628515	-0.33320231865869104
65	52.20180496008915	...	-11.788640384763386	-2.7696489462997844
66	2094.8434487575737	...	-11.805766376914509	-2.8146351429715355

Length = 66 rows

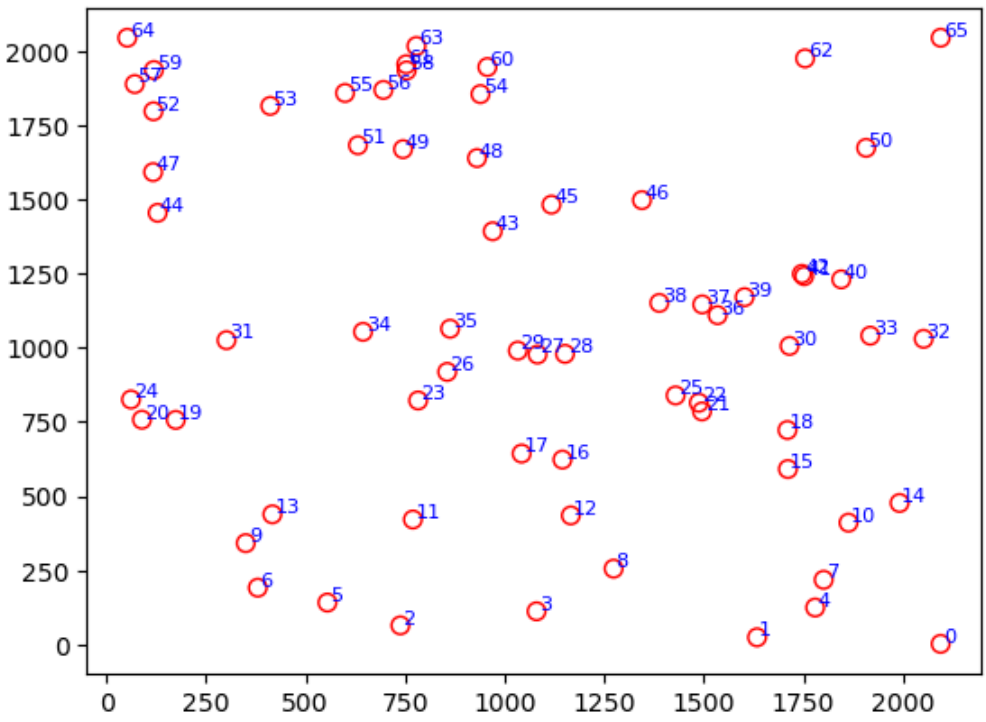


In [208…

```
if sources is not None:
    plt.scatter(sources["xcentroid"], sources["ycentroid"],
                s=50, facecolors="none", edgecolors="red")

    for i, (x, y) in enumerate(zip(sources["xcentroid"], sources["ycentroid"])):
        plt.text(x + 6, y + 6, str(i), color="blue", fontsize=8)

plt.savefig("Detected sources in NN Ser image with indices")
plt.show()
```



In [232…

```
plt.figure(figsize=(8,8))

# Proper grayscale image (fixes yellow issue)
plt.imshow(image, cmap='gray',
           vmin=np.percentile(image, 5),
           vmax=np.percentile(image, 99))

if sources is not None:
    x = sources["xcentroid"]
    y = sources["ycentroid"]

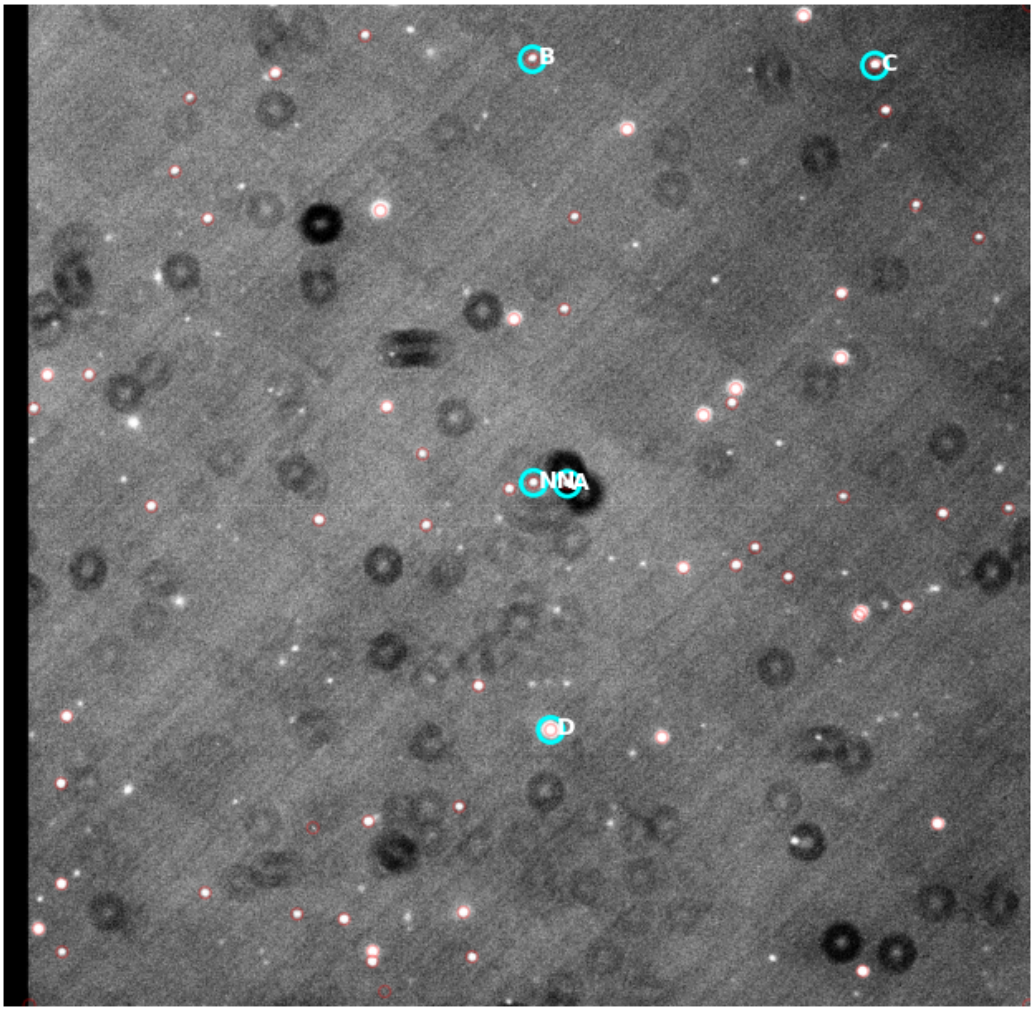
    # All detections (subtle)
    plt.scatter(x, y, s=25, facecolors="none",
                edgecolors="red", alpha=0.3)

    # Your selected stars
    selected = {
        "NN": 27,
        "A": 28,
        "B": 3,
        "C": 4,
        "D": 45
    }

    for label, idx in selected.items():
        plt.scatter(x[idx], y[idx], s=120,
                    edgecolors="cyan", facecolors="none", linewidths=2)

        plt.text(x[idx] + 10, y[idx] + 10, label,
                 color="white", fontsize=10, weight="bold")

plt.axis("off")
plt.savefig("nn_identification.pdf", bbox_inches='tight', dpi=300)
plt.show()
```



Selected Stars ZP and magnitudes computed D not Used due to bad Zero point

In [230…

```
idx_NN = 27
idx_A  = 28
idx_B  = 3
idx_C  = 4
idx_D  = 45
```

In [241…

```
m_B = 15.1
m_C = 13.7
m_D = 13.7


# --- positions ---
x = np.array(sources["xcentroid"])
y = np.array(sources["ycentroid"])

positions = {
    "NN": (x[idx_NN], y[idx_NN]),
```

```
    "B":  (x[idx_B],  y[idx_B]),
    "C":  (x[idx_C],  y[idx_C]),
    "D":  (x[idx_D],  y[idx_D]),
    'A':  (x[idx_A],  y[idx_A]),}

# --- aperture photometry ---
def get_flux(image, x0, y0):
    ap = CircularAperture([(x0, y0)], r=6.0)
    an = CircularAnnulus([(x0, y0)], r_in=10.0, r_out=15.0)

    tab = aperture_photometry(image, [ap, an])

    bkg_mean = tab['aperture_sum_1'][0] / an.area
    bkg_sum  = bkg_mean * ap.area

    return tab['aperture_sum_0'][0] - bkg_sum

# --- use your reduced image ---
image = nn_reduced

# --- measure fluxes ---
flux = {}
for name, (x0, y0) in positions.items():
    flux[name] = get_flux(image, x0, y0)

# --- instrumental magnitudes ---
mag_inst = {}
for name in flux:
    if flux[name] > 0:
        mag_inst[name] = -2.5 * np.log10(flux[name])
    else:
        mag_inst[name] = np.nan

print("----- INSTRUMENTAL MAGNITUDES -----")
for k in mag_inst:
    print(f"{k}: {mag_inst[k]:.4f}")

# --- zeropoints from B, C, D ---
zp_B = m_B - mag_inst["B"]
zp_C = m_C - mag_inst["C"]
zp_D = m_D - mag_inst["D"]

zp_vals = np.array([zp_B, zp_C])

ZP = np.nanmean(zp_vals)
ZP_err = np.nanstd(zp_vals)

print("\n----- ZEROPOINTS -----")
print(f"B: {zp_B:.4f}")
print(f"C: {zp_C:.4f}")
print(f"D: {zp_D:.4f}")
print(f"Mean ZP = {ZP:.4f} ± {ZP_err:.4f}")

# --- calibrated NN magnitude ---
mag_NN = mag_inst["NN"] + ZP
mag_A = mag_inst["A"] + ZP
print("\n----- FINAL RESULT -----")
print(f"NN Ser magnitude = {mag_NN:.4f}")
print(f'A mag = {mag_A:.4f}')
```

```
----- INSTRUMENTAL MAGNITUDES -----
NN: -11.1965
B: -11.1140
C: -12.5348
D: -15.5460
A: -12.4179

----- ZEROPOINTS -----
B: 26.2140
C: 26.2348
D: 29.2460
Mean ZP = 26.2244 ± 0.0104

----- FINAL RESULT -----
NN Ser magnitude = 15.0279
A mag = 13.8065
```

In [239…

```python
from astropy.time import Time

def get_jd(header, fallback_index=None):
    # best option for these files: combine DATE-OBS and UT
    if "DATE-OBS" in header and "UT" in header:
        date_str = str(header["DATE-OBS"]).strip()
        ut_str = str(header["UT"]).strip()
        t = Time(f"{date_str}T{ut_str}", format="isot", scale="utc")
        return t.jd

    # fallback options
    if "DATE-OBS" in header:
        try:
            t = Time(str(header["DATE-OBS"]).strip(), format="isot", scale="utc"
            return t.jd
        except:
            pass

    if "MJD-OBS" in header:
        return float(header["MJD-OBS"]) + 2400000.5

    if "JD" in header:
        return float(header["JD"])

    if fallback_index is not None:
        return float(fallback_index)

    return np.nan
```

Compute Lightcurve

In [242…

```python
import os

def get_jd(header, fallback_index=None):
    if "DATE-OBS" in header and "UT" in header:
        date_str = str(header["DATE-OBS"]).strip()
        ut_str = str(header["UT"]).strip()
        t = Time(f"{date_str}T{ut_str}", format="isot", scale="utc")
        return t.jd
    if "DATE-OBS" in header:
        try:
```

```
            t = Time(str(header["DATE-OBS"]).strip(), format="isot", scale="utc"
            return t.jd
        except:
            pass
    if "MJD-OBS" in header:
        return float(header["MJD-OBS"]) + 2400000.5
    if "JD" in header:
        return float(header["JD"])
    if fallback_index is not None:
        return float(fallback_index)
    return np.nan

times = []
nn_mags = []
nn_errs = []
zp_vals_all = []

print("----- LIGHTCURVE RESULTS -----")

for i, fname in enumerate(nn_files_clean):
    with fits.open(fname) as hdul:
        header = hdul[0].header
        image = hdul[0].data.astype(float)

    image_red = image - masterbias

    flux_NN = get_flux(image_red, *ref_pos["NN"])
    flux_B  = get_flux(image_red, *ref_pos["B"])
    flux_C  = get_flux(image_red, *ref_pos["C"])

    if flux_B <= 0 or flux_C <= 0:
        continue

    mag_B_inst = -2.5 * np.log10(flux_B)
    mag_C_inst = -2.5 * np.log10(flux_C)

    zp_B = m_B - mag_B_inst
    zp_C = m_C - mag_C_inst

    ZP = np.mean([zp_B, zp_C])
    ZP_err = np.std([zp_B, zp_C])

    if flux_NN > 0:
        mag_NN_inst = -2.5 * np.log10(flux_NN)
        mag_NN = mag_NN_inst + ZP
    else:
        mag_NN = np.nan

    jd = get_jd(header, fallback_index=i)

    times.append(jd)
    nn_mags.append(mag_NN)
    nn_errs.append(ZP_err)
    zp_vals_all.append(ZP)

    print(f"{os.path.basename(fname)}")
    print(f"  time = {jd:.8f}")
    print(f"  Mean ZP = {ZP:.4f} ± {ZP_err:.4f}")
    print(f"  NN mag = {mag_NN:.4f}")
    print("")
```

```
times = np.array(times)
nn_mags = np.array(nn_mags)
nn_errs = np.array(nn_errs)
zp_vals_all = np.array(zp_vals_all)

order = np.argsort(times)
times = times[order]
nn_mags = nn_mags[order]
nn_errs = nn_errs[order]
zp_vals_all = zp_vals_all[order]

times_rel = (times - times[0]) * 24  # hours since first image

print("----- SUMMARY -----")
print(f"Images used = {len(times)}")
print(f"Mean zeropoint = {np.nanmean(zp_vals_all):.4f}")
print(f"Zeropoint scatter = {np.nanstd(zp_vals_all):.4f}")
print(f"Mean NN magnitude = {np.nanmean(nn_mags):.4f}")

plt.figure(figsize=(8,5))
plt.errorbar(times_rel, nn_mags, yerr=nn_errs, fmt='o', capsize=3)
plt.gca().invert_yaxis()
plt.xlabel("Time since first image (hours)")
plt.ylabel("Magnitude")
plt.savefig('lightcurve.pdf')
plt.tight_layout()
plt.show()
```

```
----- LIGHTCURVE RESULTS -----
O120308_0175.fits
  time = 2455995.62067020
  Mean ZP = 26.4327 ± 0.2187
  NN mag = 15.2362

O120308_0176.fits
  time = 2455995.62165362
  Mean ZP = 26.4145 ± 0.2333
  NN mag = 15.1753

O120308_0177.fits
  time = 2455995.62264553
  Mean ZP = 26.3803 ± 0.2103
  NN mag = 15.2347

O120308_0178.fits
  time = 2455995.62364182
  Mean ZP = 26.2814 ± 0.2227
  NN mag = 15.2507

O120308_0179.fits
  time = 2455995.62560328
  Mean ZP = 26.9262 ± 0.2200
  NN mag = 15.2475

O120308_0180.fits
  time = 2455995.62694552
  Mean ZP = 26.9288 ± 0.2180
  NN mag = 15.2454

O120308_0181.fits
  time = 2455995.62828418
  Mean ZP = 26.8902 ± 0.2313
  NN mag = 15.2212

O120308_0182.fits
  time = 2455995.62962331
  Mean ZP = 26.8340 ± 0.2168
  NN mag = 15.2342

O120308_0183.fits
  time = 2455995.63096122
  Mean ZP = 26.8021 ± 0.2149
  NN mag = 15.2357

O120308_0184.fits
  time = 2455995.63230128
  Mean ZP = 26.9123 ± 0.2294
  NN mag = 15.2230

O120308_0185.fits
  time = 2455995.63364172
  Mean ZP = 26.9048 ± 0.2148
  NN mag = 15.2609

O120308_0186.fits
  time = 2455995.63498478
  Mean ZP = 26.8739 ± 0.2298
  NN mag = 15.2577
```

```
O120308_0187.fits
  time = 2455995.63632656
  Mean ZP = 26.8942 ± 0.2189
  NN mag = 15.2424

O120308_0188.fits
  time = 2455995.63766591
  Mean ZP = 26.9150 ± 0.2169
  NN mag = 15.2378

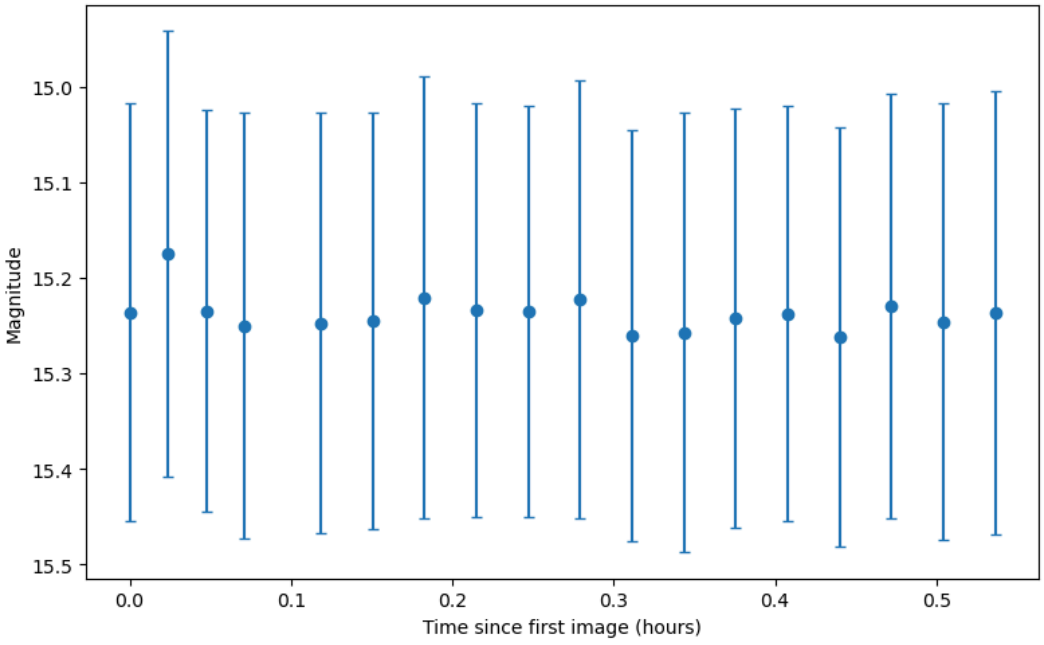
O120308_0189.fits
  time = 2455995.63900491
  Mean ZP = 26.9492 ± 0.2191
  NN mag = 15.2619

O120308_0190.fits
  time = 2455995.64034534
  Mean ZP = 26.9658 ± 0.2222
  NN mag = 15.2299

O120308_0191.fits
  time = 2455995.64169255
  Mean ZP = 26.9570 ± 0.2289
  NN mag = 15.2462

O120308_0192.fits
  time = 2455995.64302566
  Mean ZP = 26.9140 ± 0.2319
  NN mag = 15.2372

----- SUMMARY -----
Images used = 18
Mean zeropoint = 26.7876
Zeropoint scatter = 0.2244
Mean NN magnitude = 15.2377
```



TESS

In [253…

```
from astropy.timeseries import LombScargle

# --- load TESS data robustly ---
tess = pd.read_csv(
    "data/Astro_analysis_data/NN_Ser_TESS.csv",
    sep=r"\s+|,",
    engine="python",
    comment="#",
    header=None
)

# keep only first 3 columns
tess = tess.iloc[:, :3].copy()
tess.columns = ["time", "flux", "flux_err"]

# force numeric and drop bad rows
tess["time"] = pd.to_numeric(tess["time"], errors="coerce")
tess["flux"] = pd.to_numeric(tess["flux"], errors="coerce")
tess["flux_err"] = pd.to_numeric(tess["flux_err"], errors="coerce")
tess = tess.dropna()

# convert to numpy arrays
time = tess["time"].to_numpy(dtype=float)
flux = tess["flux"].to_numpy(dtype=float)
flux_err = tess["flux_err"].to_numpy(dtype=float)

# remove non-physical rows
good = np.isfinite(time) & np.isfinite(flux) & np.isfinite(flux_err) & (flux_err
time = time[good]
flux = flux[good]
flux_err = flux_err[good]

print(tess.head())
print(f"Loaded {len(time)} good rows")

# normalize flux
flux_norm = flux / np.nanmedian(flux)

# --- Lomb-Scargle periodogram ---
frequency, power = LombScargle(time, flux_norm, flux_err).autopower()

best_frequency = frequency[np.argmax(power)]
best_period = 1.0 / best_frequency

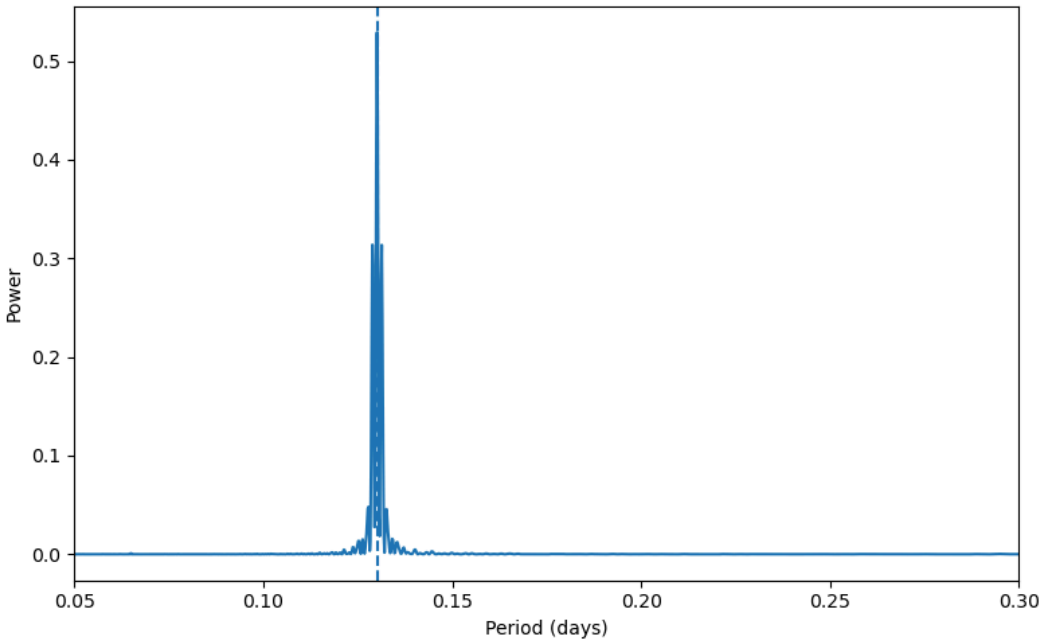
print(f"Best frequency = {best_frequency:.8f} cycles/day")
print(f"Best period = {best_period:.8f} days")
print(f"Best period = {best_period*24:.5f} hours")
print(f"Best period = {best_period*24*60:.2f} minutes")

# --- plot periodogram ---
period = 1.0 / frequency

plt.figure(figsize=(8, 5))
plt.plot(period, power)
plt.axvline(best_period, linestyle="--")
```

```
plt.xlabel("Period (days)")
plt.ylabel("Power")
plt.xlim(0.05, 0.3)
plt.tight_layout()
plt.savefig("lomb.pdf")
plt.show()
```

```
              time       flux   flux_err
23816  2698.459048  48.230282  16.322626
23817  2698.459279  32.870064  16.324450
23818  2698.459511  37.281643  16.333122
23819  2698.459742  54.776100  16.322741
23820  2698.459974  51.043427  16.349316
Loaded 46606 good rows
Best frequency = 7.68829420 cycles/day
Best period = 0.13006786 days
Best period = 3.12163 hours
Best period = 187.30 minutes
```



Phase Folded

In [246…

```
# --- fix normalization ---
flux_norm = flux / np.nanmedian(flux) - 1

# --- remove extreme outliers (helps a lot) ---
mask = np.isfinite(time) & np.isfinite(flux_norm) & (np.abs(flux_norm) < 0.1)
time_clean = time[mask]
flux_clean = flux_norm[mask]

# --- phase-fold ---
phase = (time_clean % best_period) / best_period

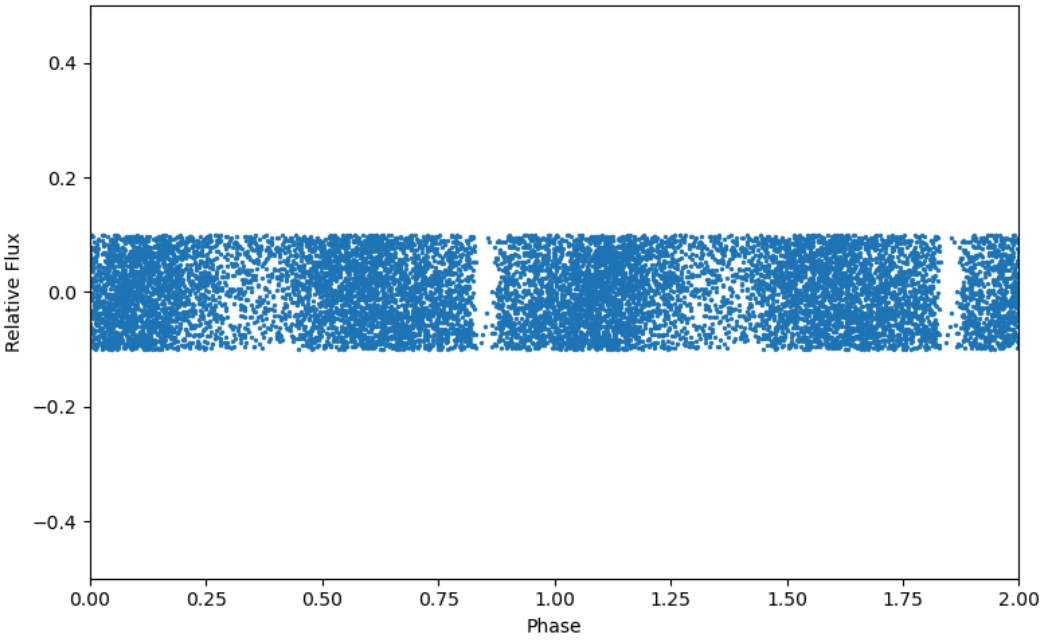
# sort
order = np.argsort(phase)
phase_sorted = phase[order]
flux_sorted = flux_clean[order]
```

```
# duplicate for 2 cycles
phase_plot = np.concatenate([phase_sorted, phase_sorted + 1])
flux_plot = np.concatenate([flux_sorted, flux_sorted])

# --- plot ---
plt.figure(figsize=(8,5))
plt.scatter(phase_plot, flux_plot, s=2)

plt.xlabel("Phase")
plt.ylabel("Relative Flux")
plt.savefig(f"phase_folded_nnser_P_{best_period:.5f}d.pdf")
plt.xlim(0, 2)
plt.ylim(-0.5, 0.5)

plt.tight_layout()
plt.show()
```



Phase folded with line to clearly signify drop off

In [249…

```
# --- normalize ---
flux_norm = flux / np.nanmedian(flux) - 1

# --- clean ---
mask = np.isfinite(time) & np.isfinite(flux_norm) & (np.abs(flux_norm) < 0.1)
time_clean = time[mask]
flux_clean = flux_norm[mask]

# --- phase ---
phase = (time_clean % best_period) / best_period

# --- binning ---
nbins = 120
bins = np.linspace(0, 1, nbins)
bin_centers = 0.5 * (bins[:-1] + bins[1:])
```

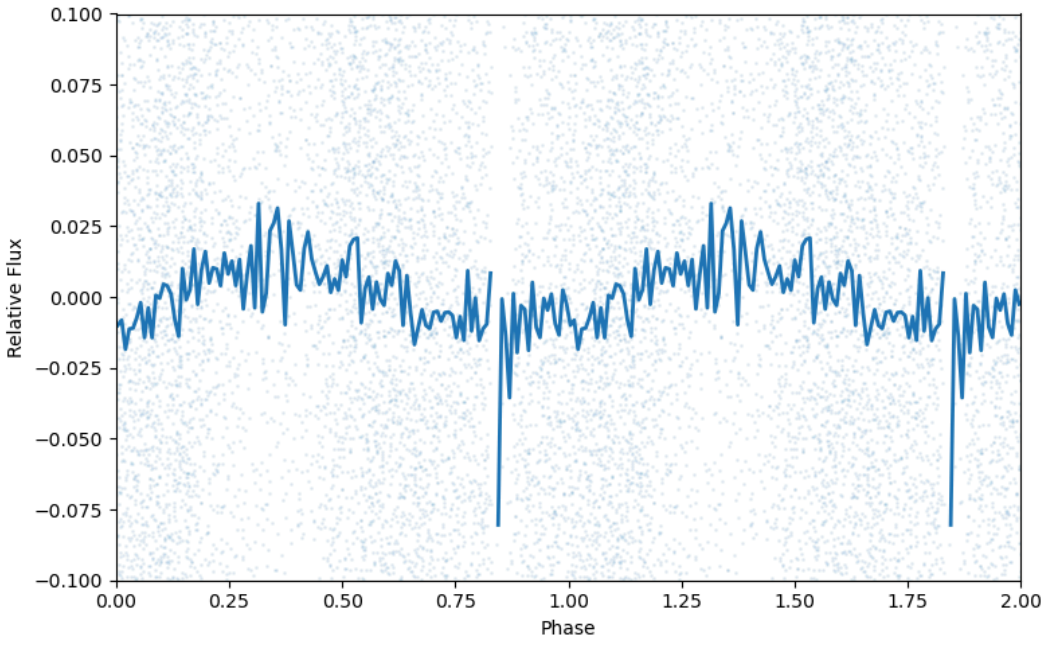
```
bin_means = np.full(nbins - 1, np.nan)

for i in range(nbins - 1):
    m = (phase >= bins[i]) & (phase < bins[i+1])
    if np.any(m):
        bin_means[i] = np.mean(flux_clean[m])

# duplicate for 2 cycles
phase_plot = np.concatenate([bin_centers, bin_centers + 1])
flux_plot = np.concatenate([bin_means, bin_means])

# --- plot ---
plt.figure(figsize=(8,5))
plt.scatter(np.concatenate([phase, phase+1]),
            np.concatenate([flux_clean, flux_clean]),
            s=1, alpha=0.08)
plt.plot(phase_plot, flux_plot, linewidth=2)

plt.xlabel("Phase")
plt.ylabel("Relative Flux")
plt.savefig('phase floded.pdf')
plt.xlim(0, 2)
plt.ylim(-0.1, 0.1)
plt.tight_layout()
plt.show()
```



In [ ]: